Journal Pre-proofs

The fate of antibiotic resistance genes in cow manure composting: shaped by temperature-controlled composting stages

Muhammad Fahad Sardar, Changxiong Zhu, Bing Geng, Hamaad Raza Ahmad, Tingting Song, Hongna Li

PII: S0960-8524(20)31677-1
DOI: https://doi.org/10.1016/j.biortech.2020.124403
Reference: BITE 124403

To appear in: *Bioresource Technology*

Received Date: 24 September 2020
Revised Date: 6 November 2020
Accepted Date: 7 November 2020

Please cite this article as: Fahad Sardar, M., Zhu, C., Geng, B., Raza Ahmad, H., Song, T., Li, H., The fate of antibiotic resistance genes in cow manure composting: shaped by temperature-controlled composting stages, *Bioresource Technology* (2020), doi: https://doi.org/10.1016/j.biortech.2020.124403

The fate of antibiotic resistance genes in cow manure composting: shaped by temperature-controlled composting stages

Authors:

Muhammad Fahad Sardar[a], Changxiong Zhu[a], Bing Geng[a], Hamaad Raza Ahmad[b], Tingting Song[a] and Hongna Li[a]*

Affiliations and addresses:

[a] *Agricultural Clean Watershed Research Group, Institute of Environment and Sustainable Development in Agriculture, Chinese Academy of Agricultural Sciences, Beijing 100081, P.R. China*

[b] *Institute of Soil and Environmental Sciences, University of Agriculture, Faisalabad, 38040, Pakistan*

* Corresponding author

Email: lihongna828@163.com (H.N. Li)

Tel: 86-010-82109561

Address: Institute of Environment and Sustainable Development in Agriculture,

Chinese Academy of Agricultural Sciences, Beijing 100081, P.R. China.

Abstract

Current work for animal manure processing is not up to the required standards and hence are not supposed to reflect the actual performance in antibiotic resistance control. As a result, this study carried out temperature-controlled aerobic composting, with sulfamethoxazole (SMX) as a typical antibiotic. The results of four different treatments demonstrated that temperature, water content, C/N ratio, EC, and pH showed no significant ($p > 0.05$) difference. Antibiotic resistance genes (ARGs)ARGs significantly decreased in the initial 10 days of the thermophilic phase, but the abundance of *sul1* and *sul2* increased greatly after 30 days. Moreover, ARGs were closely related with each other during the late stages of composting. A significant effect of composting properties, especially temperature on bacterial community, which then had a positive effect on ARGs abundances. These findings provided evidence that the standard composting was still insufficient to control antibiotic resistance.

Keywords – Programmed Temperature; Composting stages; Sulfamethoxazole; Antibiotic resistance genes; Carriers

Introduction

The prevalence of antibiotic-resistant genes in manure derived organic fertilizers has become a serious challenge to human health and environmental safety. It has been reported that 90% of antibiotics consumed by animals are excreted into the receiving environment (Zhang et al., 2020). These residual antibiotics can stimulate the rapid development of antibiotic-resistant bacteria (ARBs) and antibiotic resistant genes (ARGs) through mechanisms of mutation and horizontal genes transfer (He et al., 2020). The livestock industry produces a substantial amount of manure every year, and in China, it is approximately around 4 billion tons (Zhang et al., 2020). Respectively, this manure, as an organic fertilizer, is dumped in agricultural soils. Along with the benefits of this type of fertilization, this practice becomes the first step towards the entry of ARGs in the food chain (Guo et al., 2017). Significantly, this may pose severe constraints on food safety and human health. Consequently, according to the reports, approximately 0.1 million people die per year due to infectious diseases directly or indirectly relevant to poor agricultural practices (Zhang et al., 2015). Therefore, before it is applied to farmlands, livestock manure must be treated with promising techniques such as anaerobic digestion and aerobic composting.

Composting technology is commonly considered an economically feasible and environment-friendly technique to efficiently transform antibiotics into non-bioactive compounds and prevent the spread of ARGs and ARBs from manure to farmland soil. Different composting processes and operational procedures lead to variation and abundance of ARGs in the final product. However, aerobic composting still represents a potential avenue for ARGs and ARBs (Su et al., 2015). Furthermore, tetracyclines, macrolides, and sulfonamide resistant genes remained high in swine manure composting (Chen et al., 2019). With reference to another former study, it was reported that an abundance of five tetracycline genes has decreased during composting while two sulfonamide genes have increased, resulting in the enrichment of ARGs in the final achieved product (Wang et al., 2015). In another study, during the investigation of chicken manure composting, several ARGs associated with tetracyclines, sulfonamides, and macrolides were increased (Liu et al., 2020). Cao et al. (2020) investigated the ARGs profiles of mesophilic-thermophilic and maturation stage, and statistical analysis revealed that it was the composting stage, rather than composting material, which significantly shaped the ARGs abundance. Sun et al. (2016) reported that thermophilic condition reduced the abundance of some ARGs (*sul1* and *sul2*) more efficiently than the mesophilic condition. However, Zhang et al. (2015) indicated that resistant genes of *aadA, macB,* and *sul1* increased during the thermophilic digestion and concluded that the benefits of thermophilic digestion were not universal. However, due to the fact that some of the ARGs increased during composting; several temperature-controlled studies (30 ℃, 40 ℃, 50 ℃ and 60 ℃) have been conducted to investigate the effect of temperature on the antibiotics and the corresponding resistant determinants (Lin et al., 2017).

Previous studies have also revealed that the treatment time, temperature and operating conditions are key parameters to control ARGs prevalence and dissemination (Ma et al., 2011). Among the operating conditions, the temperature is considered as the critical parameter, and an indicator of the composting stage (Sun et al., 2016). Several studies have been conducted to study the degradation of ARGs and its influence on the microbial community, and the reported peak temperature in these experiments were 66 ℃ – 73 ℃ (Mitchell et al., 2015), 58 ℃ – 64 ℃ (Selvam et al., 2012), 55 ℃– 64 ℃ (Dolliver et al., 2008), 45 ℃– 60 ℃ (Cessna et al., 2011), 55 ℃ (Storteboom et al., 2007), and 43 ℃– 45 ℃ (Ho et al., 2013). In general, fates of the antibiotics including florfenicol, tylosin, sulfamethazine and chlortetracycline were investigated, but significant discrepancies between temperature conditions and ARGs removal existed between these studies. These results revealed the complexity of ARGs behavior during key composting stages and process conditions applied in different studies. Thus, it can be concluded that current work for animal manure processing are not up to the required standards (Smith et al., 2016) and hence are not supposed to reflect the actual performance in antibiotic resistance control. In general, the composting technique can only prove its effectiveness against pathogens, weeds, ARBs and ARGs via fulfilling the standard requirements (above 55 ℃ for ≥ 3 days). For vector attraction reduction, the temperature above 40 °C must be maintained for 14 days with an average of 45 °C as specified by US-EPA (40 CFR 60 Part 503) (Qian et al., 2016).

On the whole, the role of composting stages and the rebounding of ARGs during each stage was still a question. To investigate the actual variation of antibiotic resistance, a comprehensive temperature-programmed composting process was explored, and the temperature conditions were optimized to assure the minimum temperature required for each composting stage (Wang et al., 2015). Despite a large number of temperature-controlled studies, this experiment possesses a novel approach to investigate the differences in abundances of ARGs, MGEs, bacterial community structure and antibiotic degradation.

It was hypothesized that the comprehensive investigation of different stages of composting with artificially induced temperature phases might help to elucidate the optimal conditions for removing residual ARGs from manure derived fertilizer. To ensure reliability and repeatability of results, this study employed sulfamethoxazole (SMX), which is typically used to treat infectious diseases in livestock, was selected as a representative class of the sulfonamide drugs. By investigating the profile variation of sulfonamide associated resistant genes, *intI1*, degradation of SMX, the succession of bacterial communities, and the patterns of physicochemical parameters during cow manure composting, this study aimed to 1) determine the effect of temperature-programmed composting stages on ARGs profile; 2) uncover the microbial succession of communities based on advanced molecular techniques, and 3) optimize the aerobic composting conditions for efficient control of ARGs and their rebounding. This study provides a better understanding to mitigate the public risk of ARGs, safe production of organic fertilizer, and management of composted manure with a high abundance of ARGs.

2. Materials and Methods

2.1 Composting experiment and physicochemical analysis

Fresh cow manure and dried corn straw were collected from a local farm in the Shunyi district, Beijing, China. The fresh manure and corn straw had a water content of 88% and 6%, pH of 7.7 and 5.9, total carbon content (dry weight basis) of 38% and 72%, and total nitrogen content (dry weight basis) of 1.88% and 0.75%, respectively. Fresh cow manure was placed on plastic sheets and was air-dried (20 – 25 °C for c. 5 days). The dried manure and straw were mechanically crushed to obtain a uniform size and then sieved through 0.5 – 1 cm mesh.

The aerobic composting experiment was performed in a biochemical incubator for a period of 45 days, where the temperature was monitored and controlled at 8:00 hours for each interval. The compost devices comprised of 12 rectangular high-density polyethylene reactors (height 50 cm and width 40 cm). Four holes at the top (2 x 2 cm) and six holes of the same size at each side of the reactor were made to facilitate aeration. Composting mixtures were prepared by mixing cow manure and corn straw to achieve the carbon/nitrogen (C/N) ratio of 30:1 and were adjusted to the moisture content of 65%. A stock solution of SMX (4-amino-*N*-(5-methylisoxazol-3-yl)-benzene sulfonamide) (1000 mg L^{-1}) was prepared in 99.9% hydrochloric acid analytical grade and was diluted with an appropriate volume of sterile water. All other chemicals, of at least analytical grade, were purchased from Beijing Chemical Reagents Co. (Beijing, China). Subsequently, the diluted SMX solution was mixed thoroughly with the composting material as mentioned below: Composting without the addition of antibiotic was labelled as (T0); The treatments spiked with SMX 25, 50, and 100 mg kg^{-1} were represented by T1, T2, and T3 respectively. Furthermore, each treatment had three replicates. The reactors were placed in the biochemical incubator whose temperature was set as follows: 45 °C at day 1, 55 °C at day 5, 50 °C at day 11, 45 °C at day 21, 40 °C at day 31, 35 °C at day 41, and 25 °C at day 45 (storage phase) (Wang et al., 2015). Furthermore, the composting material was manually mixed two times a week to provide aeration.

The four prepared compost treatments were immediately sampled in triplicate as day 0 sample. Three samples were further collected during the thermophilic phase on day 5,10, and 20. Two samples were collected during the mesophilic phase on day 30 and 40, and one sample was collected at the compost maturity stage on day 45. Each sample was divided into two parts, where the first was stored at 4 ℃, and the second was freeze-dried and sieved through 2 mm with an ultra-centrifugal mill and stored at -80 ℃ for antibiotics determination and molecular biological analysis. Physicochemical parameters of compost samples were determined by adopting the standard procedures (Chen et al., 2019). The concentration of residual antibiotics was determined by using liquid chromatography-electrospray ionization tandem mass spectrometry (LC-ESI-MS/MS) through established procedures (Meng et al., 2017).

2.2 Molecular Biology analysis

2.2.1 DNA extraction and absolute quantitative PCR (AQ-PCR) of ARGs

The samples collected on days 0, 10, 30, and 45 were used for DNA extraction. Genomic DNA was extracted from 0.25 g of compost samples by using a DNA Isolation Kit by following the manufacturer's instructions (Omega Biotech, Norcross, GA, USA). The extracted DNA was stored in a freezer at -20 ℃ until use. End-product PCR assays were carried out to assess the most frequently detected SR genes (*sul1*, *sul2*, *dfrA1*, *dfrA7 sulA, and intI1*) and *16S* rDNA using the specific primers listed in Supplementary Material. The qPCR reaction system had a volume of 20 μL, which contained 1 μL DNA template, 10 μL of SYBER real-time PCR premixture, and 0.4 μL of each 10uM primer. The AQ-PCR analysis was performed using a Bio-Rad IQ5 system (Bio-Rad, USA). The procedure can be described in detail as follows: initial denaturation was carried out at 95 °C for 5 min, followed by 40 cycles at 95 °C for 15 s, 60 °Cfor 30 s, and a final extension at 72 °C for 32 (Song et al., 2020). Melting curve analysis was performed to counter the nonspecific amplification and visualization in 1.5% agarose gels. Sterile water was further used as the negative control in every run. The external reference method was used to calculate the copy number of ARGs, with the squared correlation coefficient (r^2) of the standard curve > 0.99 and the amplification efficiency ranging between 95% and 110%. Each gene was quantified in triplicate, and the result was calculated as the average.

2.2.3 Illumina MiSeq sequencing, sequence processing and data analysis

To characterize the presented bacterial community in the compost, high-throughput sequencing of the V3–V4 regions of the *16S* rRNA gene was carried on by Personal Bio (Shanghai, China) using the Illumina MiSeq 2500 platform. Barcoded fusion primers were added to the 338F (ACTCCTACGGGAGGCAGCAG) and 806R (GGACTACHVGGGTWTCTAAT) primers; allowing the assignment of DNA reads into specific treatment groups. PCR and thermal cycling parameters were the same as those given previously (Li et al., 2019). Operational taxonomic units were clustered with a 97% similarity cutoff using UPARSE (version 7.1, http://drive5.com/uparse/), and thus OTU representative sequence was obtained. Each OTU's representative sequence was assigned to a taxonomic level in the RDP database using the RDP classifier. To eliminate the differences caused by variations in the sequencing depth among samples, the least number of obtained sequences were picked randomly for each sample. Finally, according to the obtained results, bioinformatics analysis was carried out.

2.3 Statistical analysis

The data were presented as the mean values of three replicates, and standard deviations calculated using Microsoft Excel software (Version 2019, USA), and figures were generated using Origin software (Version 2019, USA). The relationship among ARGs was assessed by adjusting the Spearman’s and Pearson's correlation analysis using R 3.5.1. A difference in relative physicochemical parameters, the abundance of ARGs, and bacterial community were compared using the least significant difference test ($p < 0.05$). Partial least square - Path Model (PLS-PM) was conducted using Smart PLS (Version 3.3.2, USA) to explore the complex relationship network between the composting conditions, bacterial community structure, temperature, and to establish underlying relationships between ARGs and MGEs in the composting process. All the statistical analyses were performed using software IBM SPSS v19.0.

3. Results and Discussion

3.1 Changes in physicochemical parameters during manure composting

Based on the temperature program, the entire composting process was divided into three phases, namely, thermophilic (Day 1–20), mesophilic (Day 21–40), and maturation phase (Day 41–45). Fig. 1A shows that all the four treatments underwent the entire composting process, including thermophilic, mesophilic, and maturation phase, and became dark brownish and humified. The temperature of all the four treatments was followed by US-EPA standards.

Figure 1A shows that the temperature of the composting material was higher than the programmed temperature, which provided an understanding that bacterial activities and growth were not retarded. Since composting is a biological process, the microbial activity is responsible for the degradation of organic matter, and during the composting process, inherent energy is released in the form of sensible heat (Smith et al., 2016). This generation of heat during the process is a crucial parameter for completing each composting stage. In the current study, the temperature of each composting stage was artificially maintained to ensure the standard temperature requirements and biological activity reflects the temperature conditions. Therefore, the temperature was monitored; hence it was beyond the artificial conditions, which showed bacteria well adopted the induced temperature, and their metabolic activities were realistic and representatives of composting conditions (Robledo-Mahón et al., 2019). Hu et al. (2010) reported a similar trend when composting pig and poultry manure, where the application of antibiotics did not influence the temperature profiles.

During the initial phase of composting (Day 1–5), the pH decreased in T0, T1, T2, and T3, respectively (Fig. 1B). From Day 5 to Day 10, pH values of all treatments experienced an increase. During the mesophilic phase pH of almost all the treatments increased until day 40, and it was slightly stable during the maturation phase (Li et al., 2020). The analysis revealed that pH values of T0, T1, T2, and T3 during thermophilic, mesophilic, and maturation phases were statistically different from each other ($p < 0.05$).

As shown in Fig. 1C, during the thermophilic phase, the EC (mS cm^{-1}) of the material was slightly reduced initially and then increased in all the treatments. During the mesophilic stage, a gradual increase in EC was observed, which might be due to mineral salts such as phosphates and ammonium generated from the breakdown of compost material (Villa Gomez et al., 2019). An unusual observation was recorded in T1, which possibly due to higher microbial activity at a low level of supplemented antibiotic. Statistical observations showed that EC of thermophilic and mesophilic phases was significantly different ($p < 0.05$).

Water content is closely related to the dissolved organic matter and enzyme activities, thereby influencing the microbial shifts of bacterial communities (Sun et al., 2020). A rapid decrease in moisture content was noticed from Day 1–5 (thermophilic phase), which reflected higher microbial activity and high temperature during this period (Figure 1D) (Guo et al., 2012). After the initiation of the mesophilic phase (Day 20–40), a decline in moisture content was slightly inclined. After completion of the maturation phase, the moisture content was nearly 30% in all the treatments, and it fulfilled the quality of < 40% humidity criteria in the final composting product (Estrella-González et al., 2020).

According to the compost quality criteria, the C/N ratio must be between 15–20 (Cao et al., 2020). The initial C/N ratio was between 25–30, and as the entire composting process is governed by the microbial community, which was mainly responsible for attaining the quality parameter as reported above. In this experiment, the analysis showed that there was a gradual decrease in C/N ratio in all the treatments (Fig. 1E). The application of antibiotics can have little or a sharp impact on microbial activity (Qian et al., 2016). Youngquist et al. (2016) also reported that raw material and type and concentration of antibiotics might influence the microbial growth and metabolic functioning. However, this study demonstrated that the composting stages were not affected by antibiotics addition, and the final products have met the quality standards. The result indicated that the microbial process generally humifies organic matter, and the supplementation of antibiotics had no adverse effect on the total bacterial population.

Above all, these results of physicochemical parameters with four different treatments demonstrated that temperature, water content, C/N ratio, EC, and pH in T0, T1, T2, and T3 were statistically not different from each other ($p > 0.05$). However, when the experimental data was analysed according to the composting phases, the results revealed a significant difference ($p < 0.05$).

3.2 Effect of composting stages on the degradation of sulfamethoxazole

The degradation % of SMX in the control and antibiotic supplemented treatments is shown in Table 1. The results indicated that during the initial thermophilic phase (Day 5), the reduction % was increased quickly, reaching 100%, 57.3%, 57.5%, and 53.9% in T0, T1, T2, and

T3, respectively. Afterwards, on Day 10, the removal % of SMX in T1, T2, and T3 was increased to 91.2%, 83.4%, and 82.1%, respectively. These results were in agreement with the existing literature in that the degradation rate of antibiotics was concentration-dependent and that the removal rate slows down at higher concentrations (Cheng et al., 2019). Yang et al. (2016) indicated that the biodegradation of veterinary antibiotics depends on microbial activity, which is directly linked to the composting stage. Mitchell et al. (2015) reported that thermophilic temperature facilitated the degradation of antibiotics, and it has reached to ≥ 95% after 21 days of composting. Loftin et al. (2008) reported a significant positive correlation between temperature profile and transformation/degradation of several antibiotics. Although antibiotics tested in the current study were different from the abovementioned experiment, there was a similar outcome that antibiotics lost quickly during the thermophilic phase. Youngquist et al. (2016) found that the reduction of extractable antibiotics is partially dependent on abiotic conditions, and the temperature is among the key factors responsible for degradation. Later, the removal % of SMX became much slower, with the final removal of 94.6% (T1), 88.8% (T2), and 84.1% (T3) on Day 15. On Day 20, the results were consistent, and the concentration of SMX continually decreased, and the removal rate further increased to 95.0%, 92.2%, and 90.0% in T1, T2, and T3, respectively. In this research, the SMX in all the treatments had entirely degraded in the initial 25 days, and significant differences ($p < 0.05$) in degradation were observed in different composting stages. Cessna et al. (2011) conducted a study on composted feedlot manure and reported that the degradation rate of sulfonamides slows down after 36 days of processing. The degradation rate and time have varied considerably, and there exist conflicting results regarding the degradation of antibiotics. For example, (Ray et al., 2017) reported a relatively slower degradation rate after 14 days of traditional composting, and the removal rate was declined to nearly 5%. During the current study, thermophilic conditions (> 55 °C) were attained in the initial 3 days and then gradually reduced. This high temperature facilitated the degradation of antibiotics. Also, with a higher temperature and enhanced molecular collisions, the abiotic degradation mechanisms were accelerated (Wang et al. 2015). It is important to note that the current study focused on the degradation of parent antibiotic compounds and involved both biotic and abiotic processes; therefore, in addition to total extractable antibiotics, the effect of bioavailable fractions must be investigated in the future studies.

3.3 Bacterial Community

3.3.1 Bacterial richness and diversity

Apparent changes were obtained from the cow manure samples collected from control and antibiotic spiked treatments at different composting stages. Collected sequences were clustered into 467 bacterial OTUs at the most 1% of nucleotide differences. Good's coverage index (> 0.99) and refraction curve index indicated that most of the bacterial species had been detected in the compost samples (Supplementary material). A flower plot was constructed to visualize the bacterial OTUs detecting 7264 in raw material, which significantly ($p < 0.01$) were declined in all the treatments. At Day 10, OTUs numbers in T0, T1, T2, and T3 were 3801, 2920, 3317, and 1961 while at Day 30 OTUs numbers further decreased to 1995, 1712, 1656, and 1535, respectively. At Day 45, the OTUs' numbers in T0 and T1 were decreased to 1776 and 1573, respectively. During the cooling phase, the bacterial diversity (OTUs) was slightly increased in T2 and T3 (1683 and 1961 OTUs, respectively). The results showed that samples of Day 30 and Day 45 had nearly similar bacterial communities. In this regard, Cycoń et al. (2019) reported that elevated concentrations of antibiotics affected the microbial structure, and with the influence of time, a preferential outgrowth of ARBs occurred, resulting in changes to antibiotic sensitivity of the entire microbial community. Even the degradation of antibiotics was achieved in the initial 25 days, but it has been reported that the lowest concentration of antibiotics imparted resistance in the bacterial genome (Grenni et al., 2018). Therefore, the increased abundance in T2 and T3 treatments can be partially explained by the functional and genetic diversity of soil microorganisms (Reichel et al., 2015).

3.3.2. Taxa with altered relative abundance during composting stages

The variation in bacterial composition and diversity over time is shown in Figure 2. Compost samples were collected from each treatment at different stages of the process. The profile of raw material showed higher diversity than the samples collected at later stages.

The relative abundances in the raw material were Proteobacteria (38%), Bacteroidetes (24%), Firmicutes (20%), Chloroflexi (7%), and Actinobacteria (6%); which represented 95% of the total bacterial population. The profiles of bacterial diversity remarkably were changed during the course of composting. As the temperature and antibiotic concentrations declined over time, the selective pressure alleviated substantially, leading to significant changes in the bacterial community structure (Wang et al., 2015). In T0 at Day 10, Proteobacteria (28%), Bacteroidetes (8%), Firmicutes (10%), Chloroflexi (31%) and Actinobacteria (9%) represented 95% of the total bacterial population. At Day 30, Proteobacteria, Firmicutes, and Actinobacteria further decreased to 20%, 2%, and 7%, respectively. While relative abundances of Bacteroidetes and Chloroflexi increased (11% and 42%, respectively). During the cooling phase at Day 45, the relative abundances of Proteobacteria, Bacteroidetes, Firmicutes, Chloroflexi, and Actinobacteria were 21%, 10%, 2%, 40%, and 11%, respectively.

There were notable variations observed in the antibiotic supplemented treatments. In T1, the relative abundances of Proteobacteria, Bacteroidetes, Firmicutes, Chloroflexi and

Actinobacteria at Day 10 were 29%, 9%, 8%, 33%, and 7%; however, this trend was almost reversed at Day 30; 17%, 15%, 2%, 39%, and 8% respectively. While at Day 45, only slight changes were observed in Proteobacteria (18%), Bacteroidetes (14%), Firmicutes (3%), Chloroflexi (38%), and Actinobacteria (12%).

In T2, Bacteroidetes significantly were decreased at Day 10 (9%) and then increased to 14% at Day 30 and 45. At Day 10, Proteobacteria was significantly reduced to 25%. At Day 30 and Day 45, their relative abundance was further reduced to 19%. During the composting process (10, 30, and 45 days), Firmicutes were reduced from 16% to 2%. Chloroflexi and Actinobacteria showed an increasing trend and accounted for 12%–43% during the composting process.

The samples taken from T3 showed significant variations. Actinobacteria gradually increased, e.g., 7% at Day 10, 8% at Day 30, and 12% at Day 45. There was a sharp reduction observed in the abundance of Firmicutes and at the completion of composting. At Day 45, the relative abundance was 2%; moreover, it was very similar to the sample collected at Day 30. Chloroflexi emerged as a dominant phylum during the entire composting process, and it contributed 40%–35% in the total bacterial diversity. Peng et al. (2017) indicated in an experiment that Actinobacteria, Bacteroidetes, Firmicutes, and Chloroflexi were significantly enriched in antibiotic polluted soils.

3.4 Antibiotic resistant genes

3.4.1 Effect of temperature variations on the absolute abundance of ARGs

The initial absolute abundance (AA) of sulfonamide resistant genes (SRGs) was in the range of 2.76×10^{11}– 6.56×10^{8} copies/g. For *intI1*, the abundance in the initial sample was 1.38×10^{11} (Fig. 3). The raw material contained a high absolute abundance of ARGs and MGEs, and the results were in line with those reported by Cheng et al. (2019), who found that SRGs were abundant in animal manure, possible because sulfonamides were the most common antibiotics used in animal feed. At Day 10, the AA of *sul1, sul2, sulA, drfA7, and drfA1* remarkably decreased in T0 and T1 and ranged between $1.06 \times 10^{11} – 2.39 \times 10^{11}$, $1.48 \times 10^{10} – 1.84 \times 10^{10}$, $3.76 \times 10^{8} – 5.62 \times 10^{8}$, $2.02 \times 10^{8} – 8.95 \times 10^{8}$, and $2.18 \times 10^{8} – 4.95 \times 10^{8}$, respectively. Besides, at this stage, the middle and high levels of antibiotic concentration had a significant influence on ARGs (Chen et al., 2019). The *sul1* and *drfA7* genes increased in T2 and T3 and ranged between $3.33 \times 10^{11} – 3.72 \times 10^{11}$ and $1.23 \times 10^{9} – 2.43 \times 10^{9}$, respectively. Zhao et al. (2017) reported that an abundance of *sul1* had been positively correlated with sulfonamides concentration. While *sul2, sulA,* and *drfA1* had a sharp decrease in gene copy number during the initial thermophilic phase. On the basis of results, it can be attributed that the absolute abundance of each gene varied considerably in all samples, and the abundance pattern of each gene was unique (Lopatto et al., 2019). The thermophilic phase could remarkably enhance the removal of ARGs and MGEs (Duan et al., 2018).

At Day 30, *sul1, sulA, drfA7, and drfA1* further decreased in T0, T1, T2, and T3. Besides the reduction of these SRGs, *sul2* significantly increased in T1 and slightly increased in T3. The *sul2* gene was usually found in various plasmids and was positively correlated to the host of SRGs (Wu et al., 2015). Therefore, after the reduction of the temperature, the absolute abundance of *sul2* was increased. At Day 45, compared to Day 30, AA of ARGs was increased, and most of the ARGs were rebounded. The abundance of SRGs *sul1, sul2, sulA, drfA1, and drfA7 was* increased manifold and ranged between $3.99 \times 10^{10} – 2.81 \times 10^{11}$, $6.61 \times 10^{9} – 5.45 \times 10^{10}$, $4.98 \times 10^{7} – 2.98 \times 10^{8}$, $3.25 \times 10^{7} – 2.47 \times 10^{8}$, and $2.78 \times 10^{7} – 4.62 \times 10^{8}$, respectively. In the end, it was revealed that the effects of different treatments influenced the AA of ARGs ($p < 0.05$). However, regardless of the control and antibiotic supplemented treatments, there was an overall increase in the abundance of ARGs with a reduction in temperature. This goes in line with previous studies showing that some heat-tolerant ARGs were not completely removed, and due to insufficient thermophilic temperature during the conventional composting process, these ARGs were bounced back (Zhang et al., 2016). It should be noted that the abundance of total ARGs decreased rapidly during the thermophilic period, before then the rebounding, possibly because the later stage may have provided a suitable environment for regenerating the host bacteria of some ARGs (Cui et al., 2020). It was suggested that bacteria carrying SRGs were less adaptive to the thermophilic phase than to the maturation phase; probably, these bacteria were reduced initially and obviously increased in later stages (McCann et al., 2019). This proposed that the degradation of ARGs required more prolonged exposure to high temperatures. From a practical perspective, these results suggested that composting products should not be stored for a longer time due to the risk of increased ARGs abundances (Liao et al., 2019).

3.4.2 Correlation analysis between ARGs and intI1 at different composting stages

The prevalence and dissemination of ARGs are interdependent, and studies have shown that the abundance of ARGs is linked with mobile genetic elements (MGEs) (Cui et al., 2016; Zhu et al., 2019). *IntI1* is a type of integrator-integrase gene, and previous research has reported that *intI1* is the most widely distributed and abundant MGE in the environment (Zhang et al., 2020). Thus, the current study evaluated the correlations between ARGs and class one integron gene (*intI1*) during the three composting stages (Day10, 30, and 45) (Fig. 4). ARGs are often linked to MGEs in the environment, and the integrase gene *intI1,* which allows exogenous genes to be inserted in the genome, increased in abundance after temperature reduction (Lopatto et al., 2019). The correlation matrix at Day 10 indicated that *dfrA7*, *sul1,* and *sul2* had a significant positive correlation, and *sulA*, *sul1,* and *dfrA1* had a negative correlation (Fig. 4A). On the other hand, integron *intI1* acted as a proxy for the prevalence of genes, and most of the ARGs were positively correlated (e.g., *dfrA7*, *sul1,* and *sul2*). This indicated that horizontal gene transfer of ARGs by *intI1* was directly linked with the aforementioned genes. As shown in (Fig. 4B), the correlations at Day 30 were notably more robust than those at Day 10. Considerably, (e.g., *dfrA7*, *dfrA1*, s*ul1,* and *sul2*), had a significant positive correlation among each other and to that of *intI1*. At Day 45, the correlations were similar to that of at Day 30. However, the overall correlation values were weaker (Fig. 4C). The correlation coefficients of *sul2*, *drfA7,* and *intI1* at Day 45 were very less than those at Day 30. This demonstrated that composting stages influence ARGs and integrase gene relationship.

3.4.3 Relationship between ARGs and dominant bacterial genera

Spearman correlation analysis was conducted to investigate the relationship between the profiles of ARGs and dominant bacterial genera during the course of composting (Supplementary material). For the formation of heatmaps, two factors were primarily focused, i.e., composting stage and concentration of supplemented antibiotics. Several ARGs (e.g., *sul1, sul2, sulA, drfA1,* and *drfA7*) and *intI1* had multiple correlative genera, which demonstrated the diverse hosts of these ARGs. Previous research has detected many resistant strains and bacterial genera in antibiotic supplemented soils. The higher bacterial resistant strains may have resulted from antibiotic residues and the dissemination of ARGs (Zhao et al., 2017). In case of T0, at Day 10, *sul1, sul2, sulA, drfA1, drfA7,* and *intI1* had *a* significant positive correlation with 8, 3, 8, 4, 7, and 11 genera, respectively. At Day 30, total significant positive correlations were reduced from 41 to 24. On the other hand, at Day 45, total significant positive correlations were increased to 36. At the completion of composting, *Ruminiclostridium*, *Chelativorans,* *Longispora*, *Flavobacterium*, *Blastocatella,* and *Romboutsia* appeared as hosts of multiple correlative ARGs.

In T1, the correlation coefficients of abundant bacterial genera and ARGs fluctuated with the composting stages. It is noteworthy that at Day 10, the total positive correlations were 37, and at Day 45, it was significantly increased to 51. *Tagaea*, *Pseudomonas*, *Truepera*, *Blastocatella*, *Anaerobic digester metagenome*, *JTB255 marine benthic group*, *Bacillus*, *Planococcus*, *Thermopolyspora*, *Thermobacillus*, *Ruminiclostridium*, *Chelativorans, Longispora*, *Flavobacterium*, *Blastocatella,* and *Romboutsia* might be the host of various ARGs. Previous studies showed that antibiotics at a low level might positively affect the resistant bacterial population, and subsequently, ARGs can be conjugated to human pathogens, resulting in infections for humans (Zhu et al., 2019).

On the other hand, at Day 10, T2 and T3 initially showed quite similar trends. While, at Day 30, the total positive correlations were reduced twofold. At Day 45, *Blastocatella* and *Cellvbrio* were only genera in T2, which had a significant positive correlation with *sul1, sul2, sulA, drfA1,* and *drfA7*. However, in T3, *Candidatus Chloroploca*, *Truepera*, *Blastocatella*, *JTB255 marine benthic group*, *Bacillus*, *Planococcus*, *Thermobacillus*, *Ruminiclostridium*, *Chelativorans, Blastocatella,* and *Romboutsia* had significant positive correlations with ARGs and *intI1*. It has been reported that ARGs can be transferred between pathogenic and nonpathogenic microorganisms by MGEs such as *intI1*. Based on these results, it can be stated that these correlations were thought to be due to possible ARGs hosts and might possess the ability to transfer ARGs (Peng et al., 2017).

3.4.4 Interdependence of variables

To investigate the interdependence of composting temperature, bacterial profile, composting properties, MGEs, and ARGs abundance in more detail, constructed a partial least-squares path model (PLS-PM) to access the direct and indirect effects between observed constructs (Fig. 5). This was conducted to determine the impact of key variables in shaping the bacterial community and their interdependence. It was found that composting temperature had a direct effect on ARG abundances (Fig. 5). However, the link between temperature and ARGs abundances was not significant ($p > 0.05$). This contradicted the opinion that composting temperature was the only contributor to ARGs reduction (Su et al., 2015). Composting properties had a significant direct effect on the bacterial community structure and composting temperature, while the bacterial community composition had significant positive direct effects on the abundances of ARGs. Our findings were in line with the previous reports that the applied temperature in conventional composting is not sufficient to degrade ARGs and MGEs. The variation in ARGs and MGEs at different composting phases was probably caused by variation of the ARGs hosting bacteria instead (Liao et al., 2019). Crucially, MGEs were strongly linked to the ARGs, while the direct effect of bacterial community composition was also an important factor.

4. Conclusion

This study showed that the application of sulfamethoxazole at low concentrations might also enhance resistance in several bacterial genera. This work indicated that most ARGs and their host bacteria decreased in the initial thermophilic phase. Respectively, the decrease in the temperature level has led to the increase of ARGs, which might be due to the rebounding of the host-microbial community. Even the standard composting with programmed temperature was still insufficient to control antibiotic resistance, and optimization of the operational conditions according to composting stages was recommended.

Acknowledgements

This work was jointly supported by the Young Elite Scientist Sponsorship Program of BAST (Beijing Association for Science and Technology) (2019–2021), Beijing Municipal Natural Science Foundation (6192029), and Basic Research Funds for Central Research Institutes of China (BSRF201903). We thank Yanrong Zhang and Bilawal Abbasi for their help during the experiment.

Supplementary data

Supplementary data associated with this work can be found in the e-version of this paper online.

References:

1. Cao, R., Wang, J., Ben, W., Qiang, Z., 2020. The profile of antibiotic resistance genes in pig manure composting shaped by composting stage: Mesophilic-thermophilic and cooling-maturation stages. Chemosphere 250, 126181. https://doi.org/10.1016/j.chemosphere.2020.126181
2. Cessna, A.J., Larney, F.J., Kuchta, S.L., Hao, X., Entz, T., Topp, E., McAllister, T.A., 2011. Veterinary Antimicrobials in Feedlot Manure: Dissipation during Composting and Effects on Composting Processes. J. Environ. Qual. 40, 188–198. https://doi.org/10.2134/jeq2010.0079
3. Chen, C., Pankow, C.A., Oh, M., Heath, L.S., Zhang, L., Du, P., Xia, K., Pruden, A., 2019. Effect of antibiotic use and composting on antibiotic resistance gene abundance and resistome risks of soils receiving manure-derived amendments. Environ. Int. 128, 233–243. https://doi.org/10.1016/j.envint.2019.04.043

4. Chen, X., Zhao, Y., Zeng, C., Li, Y., Zhu, L., Wu, J., Chen, J., Wei, Z., 2019. Assessment contributions of physicochemical properties and bacterial community to mitigate the bioavailability of heavy metals during composting based on structural equation models. Bioresour. Technol. 289, 121657. https://doi.org/10.1016/j.biortech.2019.121657
5. Cheng, D., Feng, Y., Liu, Y., Xue, J., Li, Z., 2019. Dynamics of oxytetracycline, sulfamerazine, and ciprofloxacin and related antibiotic resistance genes during swine manure composting. J. Environ. Manage. 230, 102–109. https://doi.org/10.1016/j.jenvman.2018.09.074
6. Cui, E., Wu, Y., Zuo, Y., Chen, H., 2016. Effect of different biochars on antibiotic resistance genes and bacterial community during chicken manure composting. Bioresour. Technol. 203, 11–17. https://doi.org/10.1016/j.biortech.2015.12.030
7. Cui, P., Bai, Y., Li, X., Peng, Z., Chen, D., Wu, Z., Zhang, P., Tan, Z., Huang, K., Chen, Z., Liao, H., Zhou, S., 2020. Enhanced removal of antibiotic resistance genes and mobile genetic elements during sewage sludge composting covered with a semi-permeable membrane. J. Hazard. Mater. 396, 122738. https://doi.org/10.1016/j.jhazmat.2020.122738
8. Cycoń, M., Mrozik, A., Piotrowska-Seget, Z., 2019. Antibiotics in the Soil Environment-Degradation and Their Impact on Microbial Activity and Diversity. Front. Microbiol. 10, 338. https://doi.org/10.3389/fmicb.2019.00338
9. Dolliver, H., Gupta, S., Noll, S., 2008. Antibiotic Degradation during Manure Composting. J. Environ. Qual. 37, 1245–1253. https://doi.org/10.2134/jeq2007.0399
10. Duan, M., Gu, J., Wang, X., Li, Y., Zhang, S., Yin, Y., Zhang, R., 2018. Effects of genetically modified cotton stalks on antibiotic resistance genes, intI1, and intI2 during

pig manure composting. Ecotoxicol. Environ. Saf. 147, 637–642. https://doi.org/10.1016/j.ecoenv.2017.09.023

11. Estrella-González, M.J., Suárez-Estrella, F., Jurado, M.M., López, M.J., López-González, J.A., Siles-Castellano, A.B., Muñoz-Mérida, A., Moreno, J., 2020. Uncovering new indicators to predict stability, maturity and biodiversity of compost on an industrial scale. Bioresour. Technol. 313, 123557. https://doi.org/10.1016/j.biortech.2020.123557
12. Grenni, P., Ancona, V., Barra Caracciolo, A., 2018. Ecological effects of antibiotics on natural ecosystems: A review. Microchem. J. 136, 25–39. https://doi.org/10.1016/j.microc.2017.02.006
13. Guo, A., Gu, J., Wang, X., Zhang, R., Yin, Y., Sun, W., Tuo, X., Zhang, L., 2017. Effects of superabsorbent polymers on the abundances of antibiotic resistance genes, mobile genetic elements, and the bacterial community during swine manure composting. Bioresour. Technol. 244, 658–663. https://doi.org/10.1016/j.biortech.2017.08.016
14. Guo, R., Li, G., Jiang, T., Schuchardt, F., Chen, T., Zhao, Y., Shen, Y., 2012. Effect of aeration rate, C/N ratio and moisture content on the stability and maturity of compost. Bioresour. Technol. 112, 171–178. https://doi.org/10.1016/j.biortech.2012.02.099
15. He, Y., Yuan, Q., Mathieu, J., Stadler, L., Senehi, N., Sun, R., Alvarez, P.J.J., 2020. Antibiotic resistance genes from livestock waste: occurrence, dissemination, and treatment. npj Clean Water 3. https://doi.org/10.1038/s41545-020-0051-0
16. Ho, Y. Bin, Zakaria, M.P., Latif, P.A., Saari, N., 2013. Degradation of veterinary antibiotics and hormone during broiler manure composting. Bioresour. Technol. 131, 476–484. https://doi.org/10.1016/j.biortech.2012.12.194

17. Hu, X., Zhou, Q., Luo, Y., 2010. Occurrence and source analysis of typical veterinary antibiotics in manure, soil, vegetables and groundwater from organic vegetable bases, northern China. Environ. Pollut. 158, 2992–2998. https://doi.org/10.1016/j.envpol.2010.05.023

18. Li, B., Zhang, Z., Ma, Y., Li, Y., Zhu, C., Li, H., 2019. Electrokinetic remediation of antibiotic-polluted soil with different concentrations of tetracyclines. Environ. Sci. Pollut. Res. 26, 8212–8225. https://doi.org/10.1007/s11356-019-04294-z

19. Li, M.M., Ray, P., Teets, C., Pruden, A., Xia, K., Knowlton, K.F., 2020. Short communication: Increasing temperature and pH can facilitate reductions of cephapirin and antibiotic resistance genes in dairy manure slurries. J. Dairy Sci. 103, 2877–2882. https://doi.org/10.3168/jds.2019-17453

20. Liao, H., Friman, V.P., Geisen, S., Zhao, Q., Cui, P., Lu, X., Chen, Z., Yu, Z., Zhou, S., 2019. Horizontal gene transfer and shifts in linked bacterial community composition are associated with maintenance of antibiotic resistance genes during food waste composting. Sci. Total Environ. 660, 841–850. https://doi.org/10.1016/j.scitotenv.2018.12.353

21. Lin, H., Zhang, J., Chen, H., Wang, J., Sun, W., Zhang, X., Yang, Y., Wang, Q., Ma, J., 2017. Effect of temperature on sulfonamide antibiotics degradation, and on antibiotic resistance determinants and hosts in animal manures. Sci. Total Environ. 607–608, 725–732. https://doi.org/10.1016/j.scitotenv.2017.07.057

22. Liu, Y., Dyall-Smith, M., Marenda, M., Hu, H.W., Browning, G., Billman-Jacobe, H., 2020. Antibiotic Resistance Genes in Antibiotic-Free Chicken Farms. Antibiot. (Basel, Switzerland) 9, 120. https://doi.org/10.3390/antibiotics9030120

23. Loftin, K.A., Adams, C.D., Meyer, M.T., Surampalli, R., 2008. Effects of Ionic Strength, Temperature, and pH on Degradation of Selected Antibiotics. J. Environ. Qual. 37, 378–386. https://doi.org/10.2134/jeq2007.0230

24. Lopatto, E., Choi, J., Colina, A., Ma, L., Howe, A., Hinsa-Leasure, S., 2019. Characterizing the soil microbiome and quantifying antibiotic resistance gene dynamics in agricultural soil following swine CAFO manure application. PLoS One 14, e0220770–e0220770. https://doi.org/10.1371/journal.pone.0220770

25. Ma, Y., Wilson, C.A., Novak, J.T., Riffat, R., Aynur, S., Murthy, S., Pruden, A., 2011. Effect of Various Sludge Digestion Conditions on Sulfonamide, Macrolide, and Tetracycline Resistance Genes and Class I Integrons. Environ. Sci. Technol. 45, 7855–7861. https://doi.org/10.1021/es200827t

26. McCann, C.M., Christgen, B., Roberts, J.A., Su, J.Q., Arnold, K.E., Gray, N.D., Zhu, Y.G., Graham, D.W., 2019. Understanding drivers of antibiotic resistance genes in High Arctic soil ecosystems. Environ. Int. 125, 497–504. https://doi.org/10.1016/j.envint.2019.01.034

27. Meng, M., He, Z., Xu, Y., Wang, L., Peng, Y., Liu, X., 2017. Simultaneous extraction and determination of antibiotics in soils using a method based on quick, easy, cheap, effective, rugged, and safe extraction and liquid chromatography with tandem mass spectrometry. J. Sep. Sci. 40, 3214–3220. https://doi.org/10.1002/jssc.201700128

28. Mitchell, S.M., Ullman, J.L., Bary, A., Cogger, C.G., Teel, A.L., Watts, R.J., 2015. Antibiotic Degradation During Thermophilic Composting. Water, Air, Soil Pollut. 226. https://doi.org/10.1007/s11270-014-2288-z

29. Peng, S., Feng, Y., Wang, Y., Guo, X., Chu, H., Lin, X., 2017. Prevalence of antibiotic resistance genes in soils after continually applied with different manure for 30 years. J. Hazard. Mater. 340, 16–25. https://doi.org/10.1016/j.jhazmat.2017.06.059
30. Qian, X., Sun, W., Gu, J., Wang, X.J., Sun, J.J., Yin, Y.N., Duan, M.L., 2016. Variable effects of oxytetracycline on antibiotic resistance gene abundance and the bacterial community during aerobic composting of cow manure. J. Hazard. Mater. 315, 61–69. https://doi.org/10.1016/j.jhazmat.2016.05.002
31. Ray, P., Chen, C., Knowlton, K.F., Pruden, A., Xia, K., 2017. Fate and Effect of Antibiotics in Beef and Dairy Manure during Static and Turned Composting. J. Environ. Qual. 46, 45–54. https://doi.org/10.2134/jeq2016.07.0269
32. Reichel, R., Michelini, L., Ghisi, R., Thiele-Bruhn, S., 2015. Soil bacterial community response to sulfadiazine in the soil-root zone. J. Plant Nutr. Soil Sci. 178, 499–506. https://doi.org/10.1002/jpln.201400352
33. Robledo-Mahón, Gómez-Silván, C., Andersen, G.L., Calvo, C., Aranda, E., 2020. Assessment of bacterial and fungal communities in a full-scale thermophilic sewage sludge composting pile under a semipermeable cover. Bioresour. Technol. 298, 122550. https://doi.org/10.1016/j.biortech.2019.122550
34. Selvam, A., Xu, D., Zhao, Z., Wong, J.W.C., 2012. Fate of tetracycline, sulfonamide and fluoroquinolone resistance genes and the changes in bacterial diversity during composting of swine manure. Bioresour. Technol. 126, 383–390. https://doi.org/10.1016/j.biortech.2012.03.045

35. Smith, M.M., Aber, J.D., Rynk, R., 2016. Heat Recovery from Composting: A Comprehensive Review of System Design, Recovery Rate, and Utilization. Compost Sci. Util. 25, S11–S22. https://doi.org/10.1080/1065657x.2016.1233082

36. Song, T., Zhu, C., Xue, S., Li, B., Ye, J., Geng, B., Li, L., Fahad Sardar, M., Li, N., Feng, S., Li, H., 2020. Comparative effects of different antibiotics on antibiotic resistance during swine manure composting. Bioresour. Technol. 315, 123820. https://doi.org/10.1016/j.biortech.2020.123820

37. Storteboom, H.N., Kim, S.C., Doesken, K.C., Carlson, K.H., Davis, J.G., Pruden, A., 2007. Response of Antibiotics and Resistance Genes to High-Intensity and Low-Intensity Manure Management. J. Environ. Qual. 36, 1695–1703. https://doi.org/10.2134/jeq2007.0006

38. Su, J.Q., Wei, B., OuYang, W.Y., Huang, F.Y., Zhao, Y., Xu, H.J., Zhu, Y.G., 2015. Antibiotic Resistome and Its Association with Bacterial Communities during Sewage Sludge Composting. Environ. Sci. Technol. 49, 7356–7363. https://doi.org/10.1021/acs.est.5b01012

39. Sun, W., Qian, X., Gu, J., Wang, X.J., Duan, M.L., 2016. Mechanism and Effect of Temperature on Variations in Antibiotic Resistance Genes during Anaerobic Digestion of Dairy Manure. Sci. Rep. 6. https://doi.org/10.1038/srep30237

40. Villa Gomez, D.K., Becerra Castañeda, P., Montoya Rosales, J. de J., González Rodríguez, L.M., 2019. Anaerobic digestion of bean straw applying a fungal pre-treatment and using cow manure as co-substrate. Environ. Technol. 41, 2863–2874. https://doi.org/10.1080/09593330.2019.1587004

41. Wang, J., Ben, W., Zhang, Y., Yang, M., Qiang, Z., 2015. Effects of thermophilic composting on oxytetracycline, sulfamethazine, and their corresponding resistance genes in swine manure. Environ. Sci. Process. Impacts 17, 1654–1660. https://doi.org/10.1039/c5em00132c
42. Wu, D., Huang, Z., Yang, K., Graham, D., Xie, B., 2015. Relationships between Antibiotics and Antibiotic Resistance Gene Levels in Municipal Solid Waste Leachates in Shanghai, China. Environ. Sci. Technol. 49, 4122–4128. https://doi.org/10.1021/es506081z
43. Youngquist, C.P., Mitchell, S.M., Cogger, C.G., 2016. Fate of Antibiotics and Antibiotic Resistance during Digestion and Composting: A Review. J. Environ. Qual. 45, 537–545. https://doi.org/10.2134/jeq2015.05.0256
44. Zhang, J., Sui, Q., Tong, J., Buhe, C., Wang, R., Chen, M., Wei, Y., 2016. Sludge bio-drying: Effective to reduce both antibiotic resistance genes and mobile genetic elements. Water Res. 106, 62–70. https://doi.org/10.1016/j.watres.2016.09.055
45. Zhang, M., He, L.Y., Liu, Y.S., Zhao, J.L., Zhang, J.N., Chen, J., Zhang, Q.Q., Ying, G.G., 2020. Variation of antibiotic resistome during commercial livestock manure composting. Environ. Int. 136, 105458. https://doi.org/10.1016/j.envint.2020.105458
46. Zhang, T., Yang, Y., Pruden, A., 2015. Effect of temperature on removal of antibiotic resistance genes by anaerobic digestion of activated sludge revealed by metagenomic approach. Appl. Microbiol. Biotechnol. 99, 7771–7779. https://doi.org/10.1007/s00253-015-6688-9
47. Zhao, X., Wang, Jinhua, Zhu, L., Ge, W., Wang, Jun, 2017. Environmental analysis of typical antibiotic-resistant bacteria and ARGs in farmland soil chronically fertilized with

chicken manure. Sci. Total Environ. 593–594, 10–17. https://doi.org/10.1016/j.scitotenv.2017.03.062

48. Zhu, L., Zhao, Y., Yang, K., Chen, J., Zhou, H., Chen, X., Liu, Q., Wei, Z., 2019. Host bacterial community of MGEs determines the risk of horizontal gene transfer during composting of different animal manures. Environ. Pollut. 250, 166–174. https://doi.org/10.1016/j.envpol.2019.04.037

Figure Captions:

Fig. 1. Changes in physicochemical properties of compost, (A) Temperature, (B) Moisture, (C) EC, (D) pH, and (E) C:N ratio. Note: Bars denote standard error (n = 3).

Fig. 2. The changes of bacterial community abundance during the cow manure composting: (A) Relative abundances of the top 20 genera (B) Relative abundances of the top 20 phyla.

Fig. 3. The absolute abundance of ARGs changes on Day 0, 10, 30, and 45 of composting. (A) *intI1*, *sul2* and *sul1*; (B) *dfrA7*, *dfrA1,* and *sulA*. Note: Bars denote standard error (n = 3).

Fig. 4. Pearson's Correlation analysis among ARGs as well as the *intI1* gene (A) At Day 10 (B) At Day 30, and (C) At Day 45.

Fig. 5. Partial Least Square (multivariate statistical approach) for path analysis among ARGs, MGEs, Composting temperature, Composting properties (EC, pH, Moisture, and C:N), and Bacterial community.

Table 1. The concentration of sulfamethoxazole and its reduction (%) during the composting process

	0 day	5 day		10 day		15 day		20 day		25 day	
Treatments	Conc.	Conc.	% Reduction	Conc.	% Reduction	Conc.	% Reduction	Conc.	% Reduction	Conc.	% Reduction
T0	0.033	N. D	100%	N. D	100%	N. D	100%	N. D	100%	N. D	100%
T1	0.267	0.152	57.32%	0.088	91.22%	0.0533	94.66%	0.047	95.05%	N. D	100%
T2	0.819	0.425	57.52%	0.165	83.43%	0.112	88.85%	0.077	92.26%	N. D	100%
T3	2.283	1.052	53.93%	0.406	82.19%	0.363	84.10%	0.227	90.04%	N. D	100%

*"N. D" Not detected

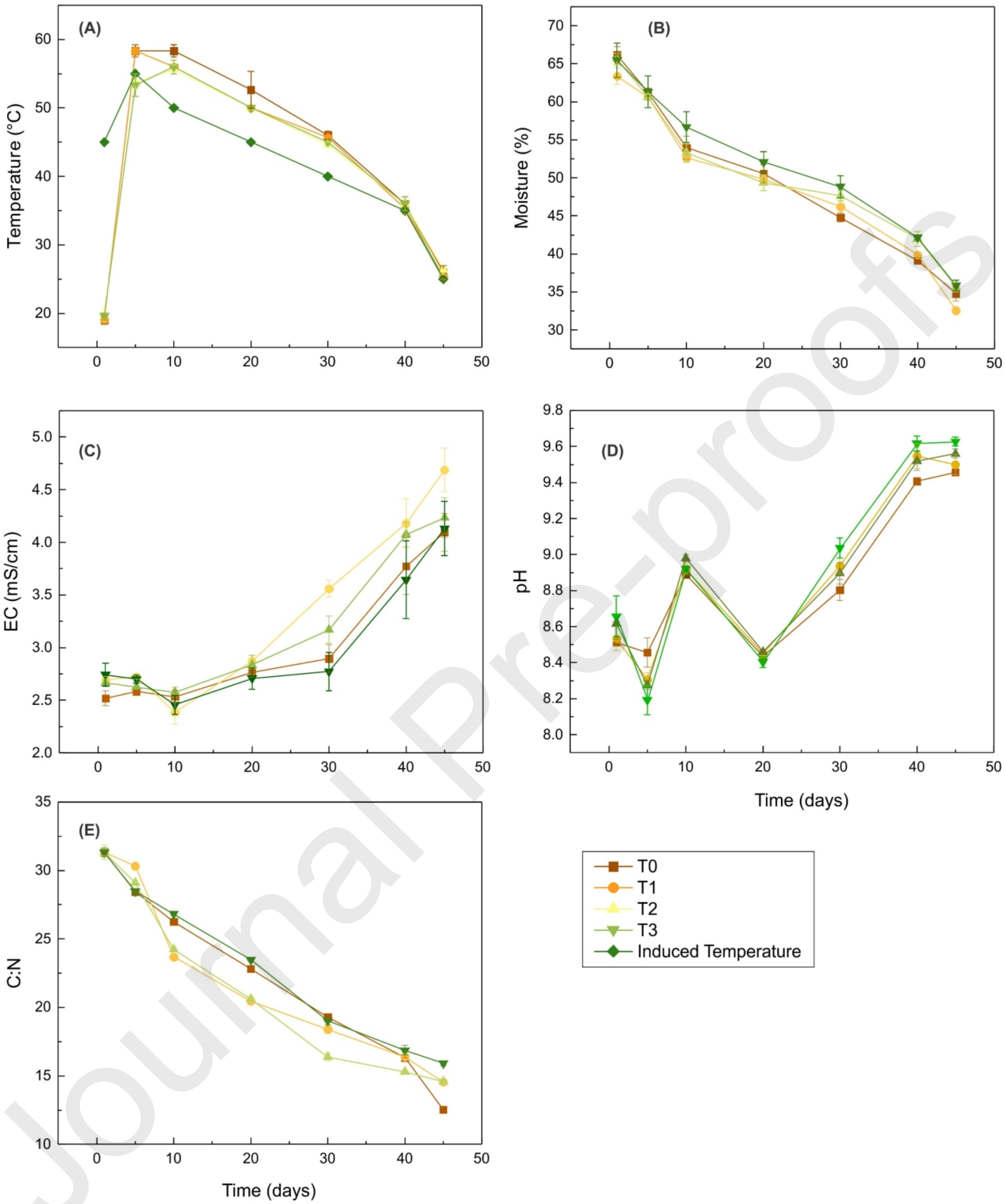
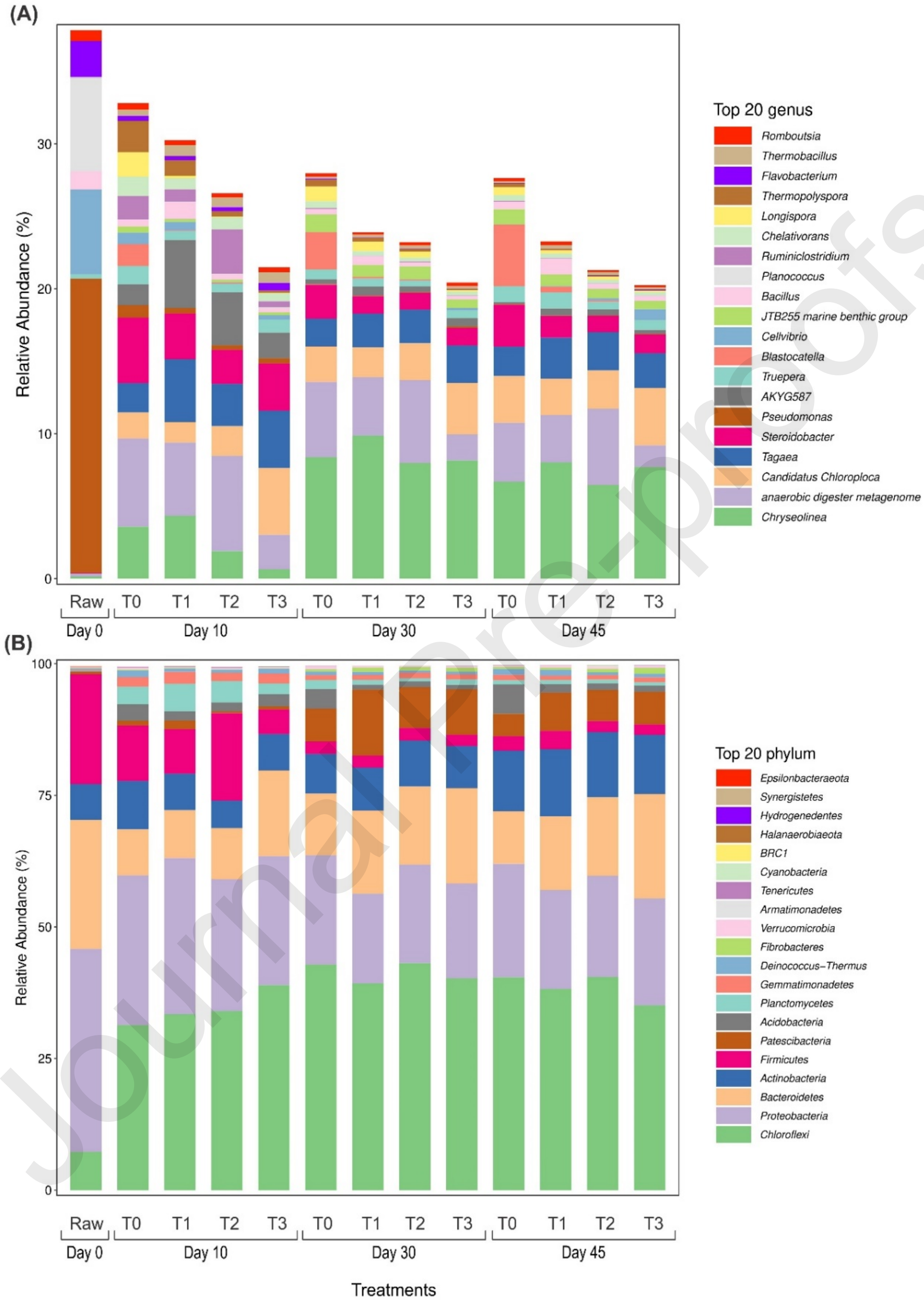

(A)
Temperature (°C)
(B)
Moisture (%)
(C)
EC (mS/cm)
(D)
pH
Time (days)
(E)
C:N
Time (days)
T0
T1
T2
T3
Induced Temperature

Fig. 1.



(A)
Relative Abundance (%)
0
10
20
30
Raw
T0
T1
T2
T3
T0
T1
T2
T3
T0
T1
T2
T3
Day 0
Day 10
Day 30
Day 45
Top 20 genus
Romboutsia
Thermobacillus
Flavobacterium
Thermopolyspora
Longispora
Chelativorans
Ruminiclostridium
Planococcus
Bacillus
JTB255 marine benthic group
Cellvibrio
Blastocatella
Truepera
AKYG587
Pseudomonas
Steroidobacter
Tagaea
Candidatus Chloroploca
anaerobic digester metagenome
Chryseolinea
(B)
Relative Abundance (%)
0
25
50
75
100
Raw
T0
T1
T2
T3
T0
T1
T2
T3
T0
T1
T2
T3
Day 0
Day 10
Day 30
Day 45
Treatments
Top 20 phylum
Epsilonbacteraeota
Synergistetes
Hydrogenedentes
Halanaerobiaeota
BRC1
Cyanobacteria
Tenericutes
Armatimonadetes
Verrucomicrobia
Fibrobacteres
Deinococcus–Thermus
Gemmatimonadetes
Planctomycetes
Acidobacteria
Patescibacteria
Firmicutes
Actinobacteria
Bacteroidetes
Proteobacteria
Chloroflexi

Fig. 2.

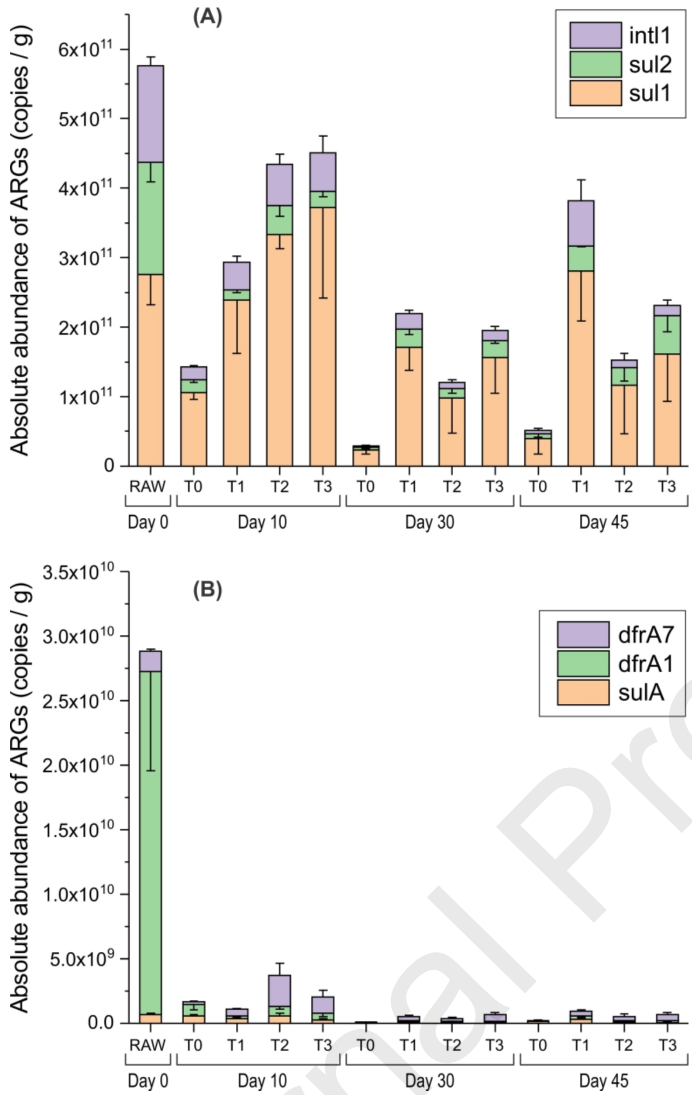

(A)
intI1
sul2
sul1
Absolute abundance of ARGs (copies / g)
6x10^11
5x10^11
4x10^11
3x10^11
2x10^11
1x10^11
0
RAW T0 T1 T2 T3 T0 T1 T2 T3 T0 T1 T2 T3
Day 0
Day 10
Day 30
Day 45
(B)
dfrA7
dfrA1
sulA
3.5x10^10
3.0x10^10
2.5x10^10
2.0x10^10
1.5x10^10
1.0x10^10
5.0x10^9
0.0

Fig. 3.

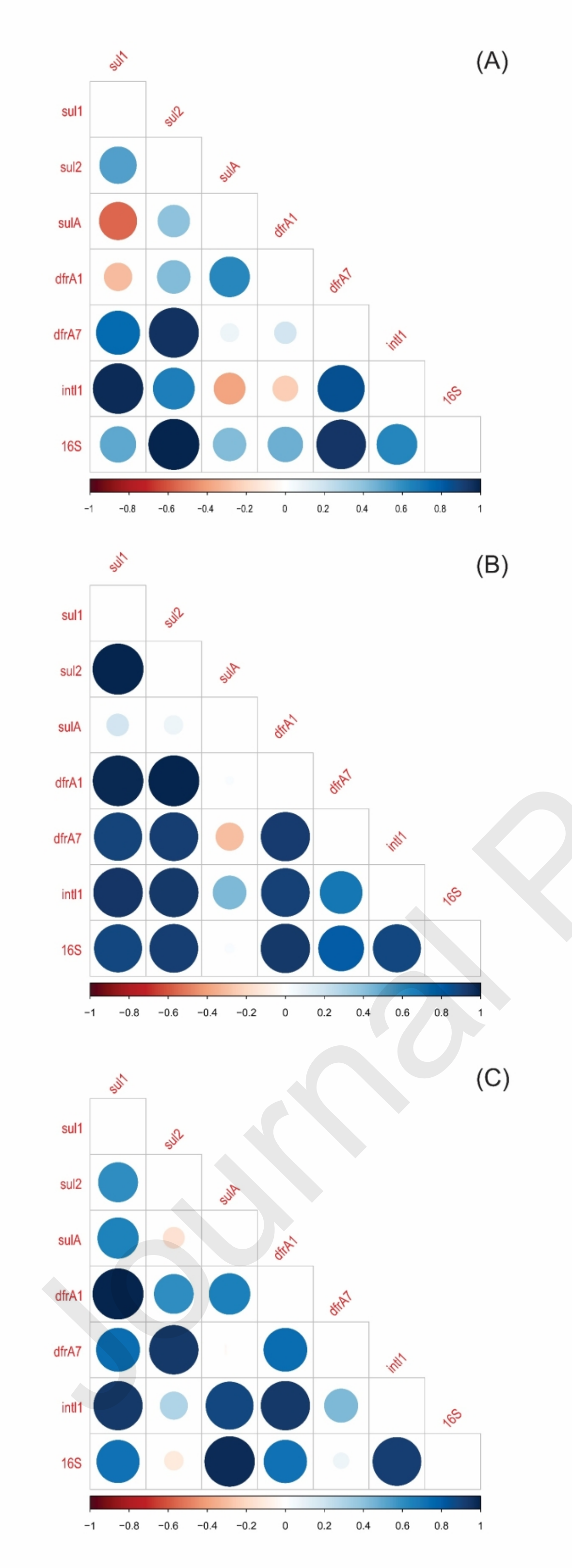

(A)
(B)
(C)
sul1
sul2
sulA
dfrA1
dfrA7
intI1
16S
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1

Fig. 4.

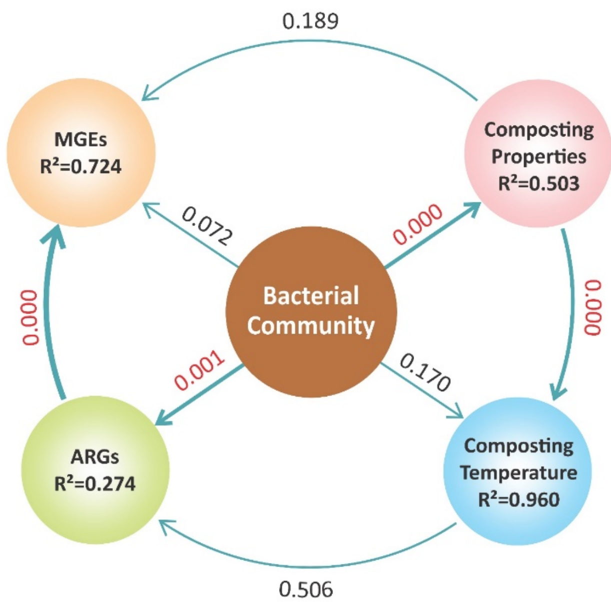

0.189
MGEs
R²=0.724
Composting
Properties
R²=0.503
0.072
0.000
Bacterial
Community
0.000
0.000
0.001
0.170
ARGs
R²=0.274
Composting
Temperature
R²=0.960
0.506

Fig. 5.

Author Contribution Statement

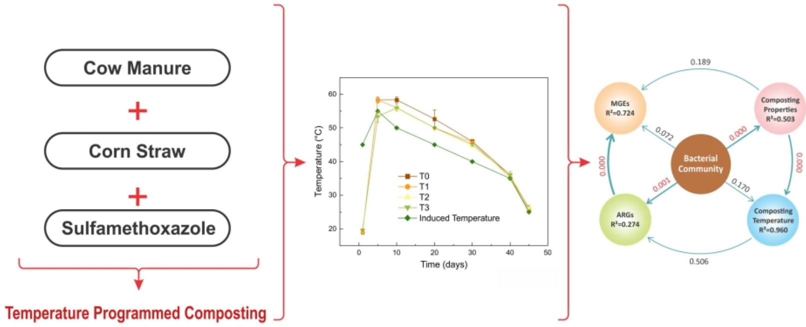
Muhammad Fahad Sardar: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing- Original draft preparation.

Changxiong Zhu and Hongna Li: Supervision, Project administration, Funding acquisition, Methodology, Resources, Review & editing and Experimental guidance.

Bing Geng: Correlated statistical analysis.

Tingting Song: Assisted with sample collection.

Hamaad Raza Ahmad: Assisted with visualization of data.


Cow Manure
+
Corn Straw
+
Sulfamethoxazole
Temperature Programmed Composting
Temperature (°C)
Time (days)
T0
T1
T2
T3
Induced Temperature
0.189
MGEs
R²=0.724
Composting
Properties
R²=0.503
0.072
0.000
Bacterial
Community
0.000
0.000
0.001
0.170
ARGs
R²=0.274
Composting
Temperature
R²=0.960
0.506

Highlights

- Actual performace of antibiotic resistance with temperature-controlled composting.
- ARGs significantly decreased initially, while *sul1* and *sul2* increased after 30 days.
- Different correlations among ARGs at thermophilic stage.
- Potential carriers of ARGs increased at low concentration of antibiotics.
- Composting properties affected bacterial community, and then ARGs abundances.